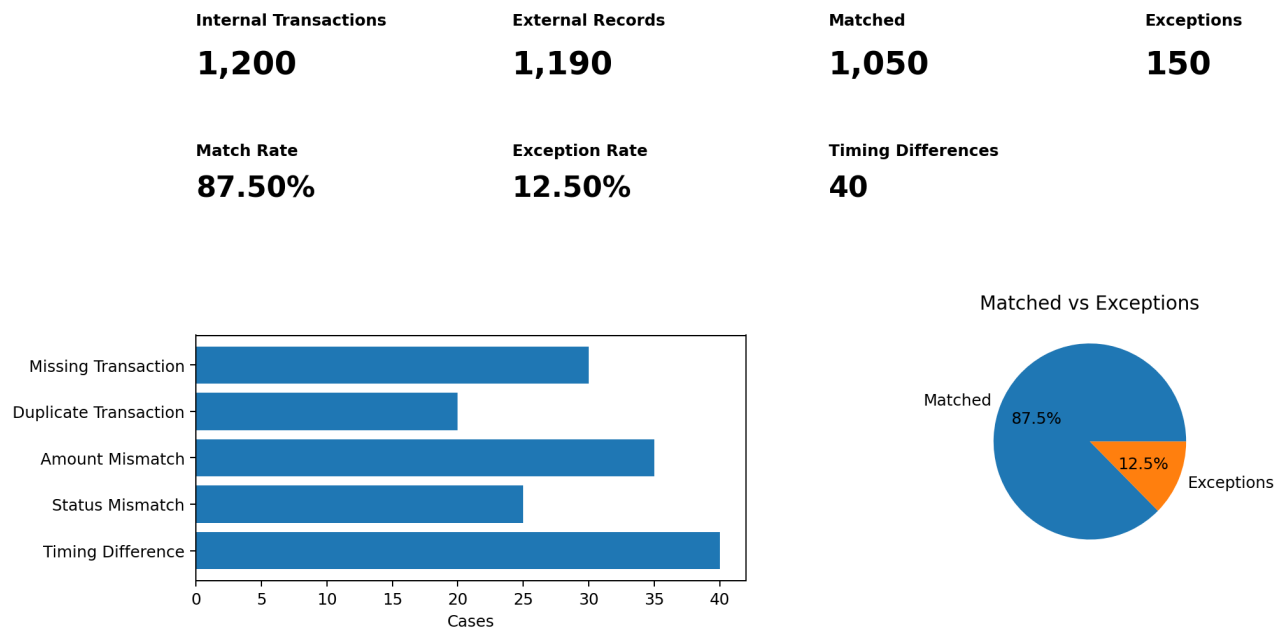


Transaction Verification & Reconciliation Analysis

Portfolio / Academic-style Case Study — Synthetic Data

TRANSACTION VERIFICATION & RECONCILIATION DASHBOARD



1. Executive Summary

This project simulates a transaction verification and reconciliation workflow between an internal transaction system and an external settlement/statement file. The analysis uses 1,200 synthetic internal transactions and 1,190 external records. The workflow demonstrates record matching, validation, exception classification, investigation tracking and KPI reporting.

2. Business Problem

Transaction-processing and finance-operations teams need reliable controls for comparing records across systems. Differences can arise from missing records, duplicate entries, amount or status mismatches, and settlement timing. A structured reconciliation process makes these differences visible and creates a clear investigation queue.

3. Objectives

Match records between two sources; validate transaction attributes; identify and classify discrepancies; maintain an exception log; calculate reconciliation KPIs; and present findings through an operational dashboard.

4. Dataset & Assumptions

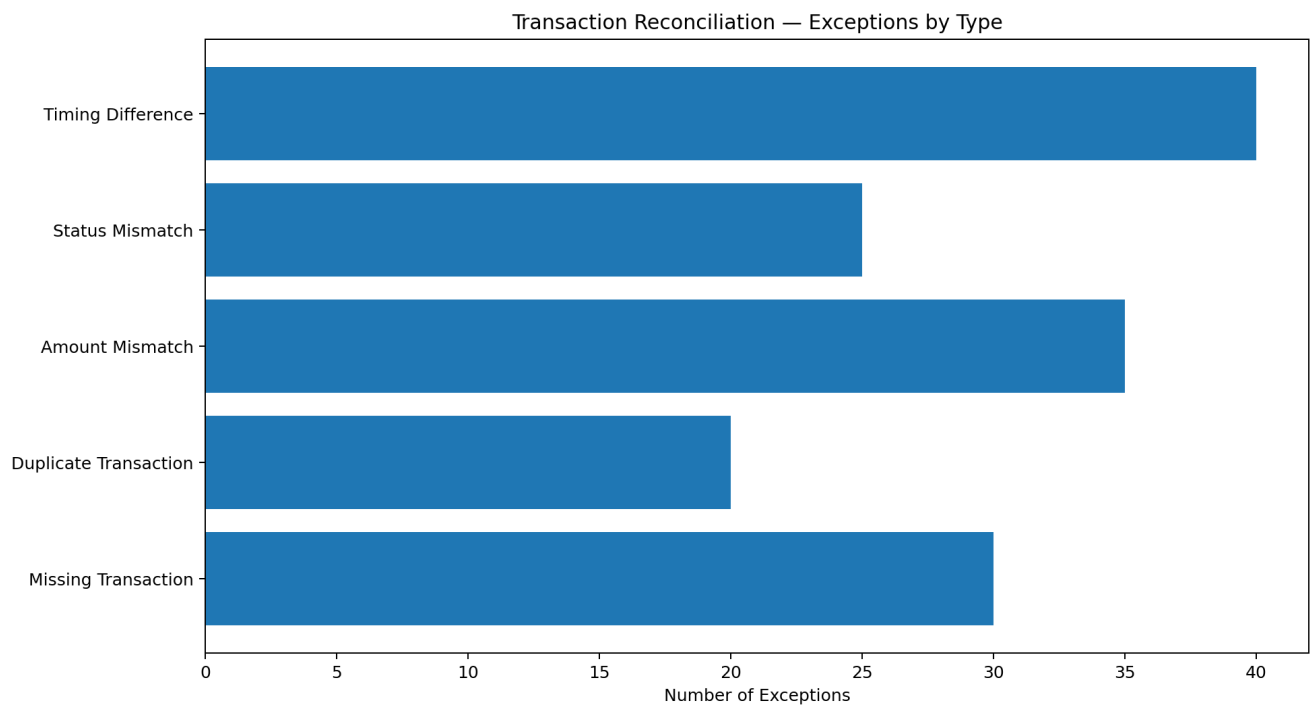
The data is entirely synthetic. The internal source contains 1,200 records and the external settlement source contains 1,190 records. Fields include transaction ID, transaction dates, settlement dates, merchant, transaction type, currency, amount, status, reference number and synthetic customer/account references. Transaction ID is the primary matching key.

5. Reconciliation Methodology

The workflow checks record existence first and then validates transaction attributes. XLOOKUP supports cross-source retrieval, IF/IFERROR supports rule-based classification, and COUNTIF/COUNTIFS and SUMIF/SUMIFS support KPI aggregation. Conditional formatting, filters and PivotTables support investigation and reporting.

6. KPI Results

KPI	Result
Internal Transactions	1,200
External Records	1,190
Matched	1,050
Exceptions	150
Match Rate	87.50%
Exception Rate	12.50%
Timing Differences	40



7. Exception Analysis

The synthetic dataset contains five controlled exception categories: 30 missing transactions, 20 duplicate transactions, 35 amount mismatches, 25 status mismatches and 40 timing differences. Timing Difference is the largest category at 26.67% of all exceptions.

8. Exception Log

The workbook contains a dedicated exception log with transaction ID, exception type, amount, date, reason/observation, priority, status and recommended action. Exceptions are not artificially marked as resolved; the project demonstrates investigation tracking rather than claiming real operational resolution.

9. Controls & Risk Perspective

The project demonstrates completeness checks, duplicate controls, amount and status validation, date/settlement checks, structured exception classification and KPI traceability to underlying records.

10. Operational Recommendations

Prioritise high-value or high-risk exceptions; review recurring timing differences; strengthen duplicate controls; validate source-system fields before reporting; track exception trends and ownership; and automate repeatable matching and validation steps where appropriate.

11. Tools & Skills Demonstrated

Microsoft Excel: XLOOKUP, IF, IFERROR, COUNTIF/COUNTIFS, SUMIF/SUMIFS, PivotTables, Conditional Formatting, Filters and Data Validation. Operations: Transaction Verification, Reconciliation, Exception Handling, Data Validation, KPI Reporting, Operational Reporting, Process Analysis and Risk & Controls Awareness. Supporting: Python/Pandas for synthetic data generation.

12. Limitations

This is a synthetic portfolio project. It contains no real customer or financial-institution data and does not represent professional reconciliation work performed for a bank, payment processor or client. Exception outcomes are simulated.

13. Conclusion

The project demonstrates an end-to-end reconciliation workflow: source comparison, validation, exception identification, investigation tracking and operational reporting. It is designed as a practical portfolio demonstration for transaction processing, finance operations, banking operations, reconciliation and back-office analytics roles.